

MATLAB

sriyukthesh100-5G6GLab/5G-6

MIMO Simulation Fix

+

Ask Gemini

matlab.mathworks.com

HOME

PLOTS

APPS

FIGURE

FORMAT

TOOLS

Search (Ctrl+Shift+Space)

Save As

Copy Figure

Print

Show Code

FILE

Labels and Annotations

Title

Subtitle

X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

X-Grid

Y-Grid

MATLAB Drive

Files

ADC_Exp1.m

ADC_Exp2a.m

ADC_Exp2b.m

DSP_Exp1_1.m

DSP_Exp1_2.m

DSP_Exp1.mat

Exp1_TestCase1.m

Exp1_TestCase2.m

Exp1_TestCase3.m

Exp1.m

SlotC_Exp2.m

Workspace

Name	Value	Size	Class
capacity	[1,2.0574,3.4...	1×7	double
snr	[0,5,10,15,20,...	1×7	double
snr_linear	[1,3.1623,10,...	1×7	double

SlotC_Exp9a.m

1 clc;

2 clear;

3 close all;

4 snr = 0:5:30;

5 snr_linear = 10.^(snr/10);

6 capacity = log2(1 + snr_linear);

7 figure;

8 plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);

9 grid on;

10 xlabel('SNR (dB)');

11 ylabel('Channel Capacity (bps/Hz)');

12 title('Channel Capacity vs SNR');

Command Window

New to MATLAB? See resources for [Getting Started](#).

>>

Figure 1

Channel Capacity vs SNR

SNR (dB)	Channel Capacity (bps/Hz)
0	1.0
5	2.0
10	3.4
15	4.7
20	6.0
25	7.4
30	8.9

33°C

Partly sunny

Search

ENG IN

10:50

13-08-2026

MATLAB

sriyukthesh100-5G6GLab/5G-6

MIMO Simulation Fix

+

Ask Gemini

matlab.mathworks.com

HOMEPLOTSAPPSFIGUREFORMATTOOLS

Search (Ctrl+Shift+Space)

Save AsCopy FigurePrintShow Code

FILE

TITLESubTitleX-LabelY-LabelZ-LabelLegendColorbarGridX-GridY-Grid

LABELS AND ANNOTATIONS

MATLAB Drive

Files

Name

ADC_Exp1.m

ADC_Exp2a.m

ADC_Exp2b.m

DSP_Exp1_1.m

DSP_Exp1_2.m

DSP_Exp1.mat

Exp1_TestCase1.m

Exp1_TestCase2.m

Exp1_TestCase3.m

Exp1.m

SlotC_Exp2.m

Workspace

Name	Value	Size	Class
capacity	[1,2.0574,3.4...	1×7	double
snr	[0,5,10,15,20,...	1×7	double
snr_linear	[1,3.1623,10,...	1×7	double

SlotC_Exp9a.m

```
1 clc;
2 clear;
3 close all;
4 snr = 0:5:30;
5 snr_linear = 10.^(snr/10);
6 capacity = log2(1 + snr_linear);
7 figure;
8 plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
9 grid on;
10 xlabel('SNR (dB)');
11 ylabel('Channel Capacity (bps/Hz)');
12 title('Channel Capacity vs SNR');
13
```

Figure 1

Channel Capacity vs SNR

SNR (dB)	Channel Capacity (bps/Hz)
0	1.0
5	2.0
10	3.3
15	4.7
20	6.3
25	7.9
30	9.6

Command Window

New to MATLAB? See resources for [Getting Started](#).

>>

33°C Partly sunny

Search

ENG IN

10:50 13-08-2026

MATLAB

sriyukthesh100-5G6GLab/5G-6G

MIMO Simulation Fix

+

Ask Gemini

matlab.mathworks.com

HOMEPLOTSAPPSFIGUREFORMATTOOLS

Search (Ctrl+Shift+Space)

Save AsCopy FigurePrintShow Code

FILE

Labels and Annotations

Figure 1

Antenna Gain for Different Antenna Array Sizes

Antenna Gain (dB)

Number of Antenna Elements

48163264

05101520

12

15

18

20

Antennas

Gain

1x5

double

1x5

double

1

2

3

4

5

6

7

8

9

10

11

12

13

clc;

clear;

close all;

antennas = [4 8 16 32 64];

gain = 10*log10(antennas);

figure

bar(antennas, gain)

grid on

% Number of Antenna Elements

% Antenna Gain (dB)

xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')

Command Window

New to MATLAB? See resources for [Getting Started](#).

>>

33°C

Partly sunny

Search

10:50

13-08-2026